

Wireless 3-D Gesture and Character Recognition

Here is a 3-D gesture recognition system that can be used for a host of applications including gesture-based user interfaces, character recognition systems, human interface devices (wireless mouse, keypad, joystick, etc) and virtual reality in 3-D gaming

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Human-machine interaction has reached the point where machines are capable of interpreting gestures of human beings. In the real world, gestures are made by hand movements and facial expressions. A gesture is detected and interpreted from the acceleration data captured due to the movement of the hand. The data is transmitted wirelessly to a receiving station where it is processed and interpreted by some application.

The gesture data is captured from a MEMS motion sensor (LIS302DL) in the form of acceleration towards x, y, and z directions using a handheld MCU MSP430F2274, and transmitted wirelessly using a 2.4GHz RF transceiver (CC2500). A receiver connected to the host PC through the USB port receives the data, which is then pre-processed and provided to the host applications. The ultra-low power mode support of MSP430F2274 makes the transmitter implementation power-efficient. The choice of components and the system design are such that the form factor is as small as possible (1x1 cm²) so it can be powered using a 30mAh button cell (CR1025). The receiver module is USB-powered.

MEMS motion sensor

The MEMS motion sensor is an ultra-compact, low-power, digital-output, 3-axis linear accelerometer consisting

of a sensing element and I²C/SPI serial interface for providing access to the external world. A proprietary process is used to create a surface micro-machined accelerometer. The technology allows creation of suspended silicon structures that are attached to the substrate at a few points called anchors, and are free to move in the direction of the sensed acceleration.

When acceleration is applied to the sensor, the roof mass displaces from its nominal position, causing an imbalance in the capacitive bridge. This imbalance is measured using charge integration in response to a voltage pulse applied to the sense capacitor.

The complete measurement chain consists of a low-noise capacitive amplifier stage followed by an analogue-to-digital converter.

The motion sensor chosen for this implementation is STMicroelectronics' IC LIS302DL. The I²C/SPI interface of this IC makes it suitable for interfacing with a microcontroller. It has a data-ready pin to indicate that a new set of acceleration data is available. This makes it suitable for use in a low-power implementation of the system.

RF transceiver

IC CC2500 from Texas Instruments has been chosen as the RF transceiver in this implementation. It is a low-cost, single-chip, 2.4GHz transceiver designed for very low-power wireless applications. The circuit is intended for the ISM (industrial, scientific and medical) and SRD (short-range device)

frequency band of 2400-2483.5 MHz.

The RF transceiver can be configured to achieve optimal performance for many different applications. Configuration is done using the SPI interface. The following key parameters can be programmed:

1. Power-down/power-up mode
 2. Crystal oscillator power-up/power-down
 3. Receive/transmit mode
 4. RF channel selection
 5. Data rate
 6. Modulation format
 7. RX channel filter bandwidth
 8. RF output power
 9. Data buffering with separate 64-byte receive and transmit FIFO
 10. Packet radio hardware support
 11. Forward error correction with interleaving
 12. Data whitening
 13. Wake-on-radio
- IC CC2500 provides extensive hardware support for packet handling, data buffering, burst transmission, clear channel assessment, link quality indication and wake-on radio.

A 2.4GHz chip antenna is used as the RF front-end with a few passive components.

MCU

The MSP430F2274 MCU from Texas Instruments has been chosen for this implementation. It is an ultra-low-power MCU with several peripherals targeted for various applications. The architecture, combined with five low-power modes (LPM0 through LPM4),